LECTURE 02 - KILLING FORM AND SEVERAL CARTAN'S CRITERIA

FANGYANG TIAN

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We continue to assume that $\mathfrak{g}$ is a finite-dimensional Lie algebra over a field F of characteristic zero in this lecture.

1. Cartan's Trace-Form Criteria

We begin with the following obvious example.

Example 1.1. *Consider the standard Borel subalgebra $\mathfrak{b}$ of $\mathfrak{gl}_n(F)$ which consists of all upper triangular matrices. Let $\mathfrak{n} := [\mathfrak{b}, \mathfrak{b}]$. Then $\mathfrak{n}$ is the ideal of $\mathfrak{b}$ which consists of all strictly upper triangular matrices. Obviously, for any $X \in \mathfrak{b}$ and $Y \in \mathfrak{n} = [\mathfrak{b}, \mathfrak{b}]$, $\mathrm{Tr}(XY) = 0$.*

Now we state the following (converse) theorem and postpone the proof of it (step by step) in Section 6.

Theorem 1.2 (Cartan's Trace-Form Criterion for Solvability)**.** *Let $\mathfrak{g}$ be a Lie subalgebra of $\mathfrak{gl}_n(F)$. Then $\mathfrak{g}$ is solvable if and only if for all $X \in \mathfrak{g}, Y \in [\mathfrak{g}, \mathfrak{g}]$, $\mathrm{Tr}(XY) = 0$.*

Recall that the radical of $\mathfrak{g}$ is the unique maximal solvable ideal of $\mathfrak{g}$ (this is the sum of all solvable ideals of $\mathfrak{g}$).

Definition 1.3. *We say a Lie algebra is **simple** if it is non-abelian and has no proper non-trivial ideals. We say a Lie algebra is **semisimple** if it has a trivial radical.*

Lemma 1.4. *A Lie algebra $\mathfrak{g}$ is simple if and only if $\mathfrak{g}$ is non-abelian and the adjoint representation of $\mathfrak{g}$ on itself is irreducible.*

Proof. Obviously, when $\mathfrak{g}$ is non-abelian, ad on $\mathfrak{g}$ has no nontrivial subrepresentation if and only if $\mathfrak{g}$ has no proper non-trivial ideals. □

Remark 1.5. *Obviously any simple Lie algebra must be semisimple, by definition.*

Corollary 1.6. *Let $\mathfrak{g}$ be a subalgebra of $\mathfrak{gl}_n(F)$. If $\mathfrak{g}$ is semisimple, then the trace form is non-degenerate.*

Proof. Define $\mathfrak{h}$ to be the radical of the trace form, i.e.

$$\mathfrak{h} := \{X \in \mathfrak{g} \,|\, \mathrm{Tr}(XY) = 0, \forall\, Y \in \mathfrak{g}\}.$$

We aim to check that $\mathfrak{h} = 0$. It is easy to check that $\mathfrak{h}$ is an ideal of $\mathfrak{g}$. Moreover, for all $X \in \mathfrak{h}$, $Y \in [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{g}$, $\mathrm{Tr}(XY) = 0$. By Theorem 1.2), $\mathfrak{h}$ is also solvable, hence contained in $\mathrm{rad}(\mathfrak{g})$. Thus,

$$\begin{aligned} \mathfrak{g} \text{ is semisimple } &\Leftrightarrow \mathrm{rad}(\mathfrak{g}) = 0 \\ &\Rightarrow \mathfrak{h} = 0 \\ &\Leftrightarrow \text{ trace form is non-degenerate.} \end{aligned}$$

□

The converse of Corollary 1.6 may fail, since $\mathfrak{h} = 0$ may not imply $\mathrm{rad}(\mathfrak{g}) = 0$. The pathology comes from the center of $\mathfrak{g}$, as indicated in the following examples.

Example 1.7. *Let $\mathfrak{a}$ be the abelian subalgebra of $\mathfrak{gl}_n(F)$ which consists of all diagonal matrices. Then $\mathfrak{a}$ is solvable. Yet the trace form is non-degenerate on $\mathfrak{a}$ (obviously).*

Example 1.8. *Let $\mathfrak{g} = \mathfrak{gl}_n(F)$. Then the center $\mathfrak{z}_\mathfrak{g}$ consists of all scalar matrices, hence* $\mathrm{rad}(\mathfrak{g})$ *is non-trivial. The Lie algebra $\mathfrak{gl}_n(F)$ is not semisimple. Yet the trace form is non-degenerate on $\mathfrak{g}$. (Let E_{ij} be the elementary matrix whose only nonzero entry lies in i-th row and j-th column. Then E_{ij} form a basis of $\mathfrak{g}$. Under the trace form, it is easy to see that the dual basis of E_{ij} is E_{ji}.)*

Remark 1.9. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra with center $\mathfrak{z}_\mathfrak{g}$. If $\mathfrak{g}/\mathfrak{z}_\mathfrak{g}$ is semisimple, then we say $\mathfrak{g}$ is reductive. (There are too many equivalent definitions/characterizations for reductive Lie algebras, we only present one here.) For reductive Lie subalgebras of $\mathfrak{gl}_n(F)$, we do need an explicit non-degenerate invariant symmetric bilinear form for explicit computations. Then the trace form does help.*

2. Cartan's Killing-Form Criteria and Corollaries

We continue to assume that $\mathfrak{g}$ is a finite-dimensional Lie algebra over F in this section. Then the adjoint representation of $\mathfrak{g}$ will 'realize' $\mathfrak{g}$ as a subalgebra of some $\mathfrak{gl}_n(F)$ ($\mathfrak{gl}(\mathfrak{g})$ to be precise) by ignoring $\mathfrak{z}_\mathfrak{g}$. Thus, the Cartan's trace-form criteria will lead to the Killing-form criteria for semisimplicity.

Definition 2.1. *For a finite dimensional Lie algebra $\mathfrak{g}$, define its* ***Killing form***

$$B(X, Y) := \mathrm{Tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y)), \quad \forall X, Y \in \mathfrak{g}.$$

Proposition 2.2. *For a finite-dimensional nilpotent Lie algebra $\mathfrak{g}$, its Killing form must be identically zero.*

Proof. Since $\mathfrak{g}$ is nilpotent, $\mathrm{ad}X$ acts nilpotently. This means that all eigenvalues of $\mathrm{ad}(X)$ are 0. Thus,

$$B(X, Y) := \mathrm{Tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y)) = 0, \quad \forall X, Y \in \mathfrak{g}.$$

□

Example 2.3. *The Killing form of the Heisenberg Lie algebra $\mathfrak{h}_n$ (and $\mathfrak{h}_n^\mathbb{C}$) vanishes identically.*

Theorem 2.4 (Cartan's Killing-Form Criterion on Solvability,[Kna02, Proposition 1.46])**.** *A finite-dimensional Lie algebra $\mathfrak{g}$ over F is solvable if and only if the Killing form vanishes identically on $\mathfrak{g} \times [\mathfrak{g}, \mathfrak{g}]$.*

Proof. Consider the adjoint representation $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$.

$$\begin{aligned} &\text{the Killing form vanishes identically on } \mathfrak{g} \times [\mathfrak{g}, \mathfrak{g}] \\ \Leftrightarrow\ &\mathrm{ad}(\mathfrak{g}) \text{ is a solvable Lie subalgebra of } \mathfrak{gl}(\mathfrak{g}) \qquad \text{(Theorem 1.2)} \\ \Leftrightarrow\ &\mathfrak{g} \text{ is solvable} \qquad \text{(Lemma 2.13 in Lecture 01).} \end{aligned}$$

□

Theorem 2.5 (Cartan's Killing-Form Criterion for Semisimplicity,[Kna02, Theorem 1.45])**.** *A finite-dimensional Lie algebra $\mathfrak{g}$ is semisimple if and only if its Killing form is non-degenerate.*

Proof. If $\mathfrak{g}$ is semisimple, then $\mathfrak{z}_{\mathfrak{g}} = 0$ and $\mathrm{ad}(\mathfrak{g}) \simeq \mathfrak{g}$ is semisimple (Lemma 2.13 in Lecture 01). Thus, by Corollary 1.6, the Killing form is non-degenerate.

Conversely, if the Killing form is non-degenerate, we need to show that $\mathrm{rad}(\mathfrak{g}) = 0$. If otherwise, then we can find a nonzero abelian ideal $\mathfrak{a} = \mathrm{rad}(\mathfrak{g})^l$ for some l. Then Take a vector space decomposition of $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$. Take any basis of $\mathfrak{a}$ and $\mathfrak{b}$ and form a basis of $\mathfrak{g}$. Under such a basis, $\mathrm{ad}(X)$ is strictly upper triangular for all $X \in \mathfrak{a}$, and $\mathrm{ad}(Y)$ is upper triangular for all $Y \in \mathfrak{g}$. This implies that $\mathrm{Tr}(\mathrm{ad}(X)\mathrm{ad}(Y)) = 0$. We get a contradiction. □

Example 2.6. *We take a basis of $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{R})$ as follows:*

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

Then

$$\mathrm{ad}(e) = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \mathrm{ad}(f) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ -1 & 0 & 0 \end{pmatrix}, \mathrm{ad}(h) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The matrix for the Killing form is

$$\begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

This is a non-degenerate form of signature $(2, 1)$. Hence $\mathfrak{sl}_2(\mathbb{R})$ is semisimple.

Remark 2.7. *One can also directly check that the Killing form $B(X, Y)$ for $\mathfrak{sl}_2(\mathbb{R})$ is equal to $4\mathrm{Tr}(X, Y)$. In fact, for $\mathfrak{sl}_n(F)$, $B(X, Y) = 2n\mathrm{Tr}(XY)$. To prove this, we need more preparations. We will only prove this claim for $F = \mathbb{R}$ or $\mathbb{C}$ in this course (the proof relies on some version of Schur's lemma and we only plan to give a proof of Schur's lemma over $\mathbb{C}$).*

Example 2.8. *Recall the basis of $\mathfrak{so}(3)$ we take in Example 1.17 in Lecture 01.*

$$X_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, X_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

One can check that

$$[X_1, X_2] = X_3, \quad [X_2, X_3] = X_1, \quad [X_3, X_1] = X_2.$$

By a direct computation,

$$\mathrm{ad}(X_1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \mathrm{ad}(X_2) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \mathrm{ad}(X_3) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The matrix of the Killing form is

$$\mathrm{diag}(-2, -2, -2),$$

which is negative definite. Hence $\mathfrak{so}(3)$ *is semisimple.*

Remark 2.9. *Negative definiteness of Killing form of* $\mathfrak{so}(3)$ *reflects the fact that* $\mathfrak{so}(3)$ *is a compact Lie algebra (a Lie algebra of a compact Lie group). Conversely, for* $\mathfrak{sl}_2(\mathbb{R})$*, the Killing form is not negative definite, hence* $\mathrm{SL}_2(\mathbb{R})$ *is noncompact. (One proof requires a (Riemannian) geometric explanation of Killing form, I hope we can mention this sometime later in the semester.)*

Example 2.10. $\mathfrak{sl}_n(F), \mathfrak{so}_{2n+1}(F), \mathfrak{sp}_{2n}(F), \mathfrak{so}_{2n}(F)$ *are simple (hence semisimple). (The proof, though can be presented elementarily, will use the idea of root space decomposition of a semisimple Lie algebra. Thus it's better to postpone the proof to later lectures.)*

3. Supplements on Killing Forms and Semisimple Lie Algberas

We continue to assume that $\mathfrak{g}$ is a finite-dimensional Lie algebra over F and B is the Killing form of $\mathfrak{g}$.

Lemma 3.1. *The Killing form* B *is invariant in the sense that*

$$B(\mathrm{ad}(H)X, Y) + B(X, \mathrm{ad}(H)Y) = 0$$

for all $H, X, Y \in \mathfrak{g}$.

Proof. This is straightforward from the definition of the Killing form. □

Lemma 3.2. *Let* $\mathfrak{h}$ *be an ideal of* $\mathfrak{g}$ *and* $B_{\mathfrak{h}}$ *be the Killing form of* $\mathfrak{h}$*. Then* $B_{\mathfrak{h}}$ *is the restriction of* B *on* $\mathfrak{h} \times \mathfrak{h}$*. (Homework)*

Lemma 3.3. *For a semisimple Lie algebra* $\mathfrak{g}$*, the orthogonal complement (w.r.t the Killing form) of an ideal* $\mathfrak{a}$ *is still an ideal and* $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$.

Proof. Set $\mathfrak{b} := \mathfrak{a} \cap \mathfrak{a}^\perp$. Then $\mathfrak{b}$ is an ideal of $\mathfrak{g}$. By Lemma 3.2, the Killing form $B_{\mathfrak{b}}$ of $\mathfrak{b}$ is the restriction of B on $\mathfrak{b}$. Hence $B_{\mathfrak{b}}$ vanishes identically. By Cartan's criterion (Theorem 2.4), $\mathfrak{b}$ must be solvable, hence must be zero as $\mathfrak{g}$ is semisimple. To check $\mathfrak{a}^\perp$ is an ideal, we note that for any $X \in \mathfrak{a}, Y \in \mathfrak{a}^\perp$ and $H \in \mathfrak{g}$,

$$B(X, \mathrm{ad}(H)Y) = -B(\mathrm{ad}(H)X, Y) = 0.$$

Hence $\mathfrak{a}^\perp$ is $\mathrm{ad}(\mathfrak{g})$ stable. □

Lemma 3.4. *For a simple Lie algebra* $\mathfrak{g}$*,* $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

Proof. Note that $[\mathfrak{g}, \mathfrak{g}]$ is a nonzero ($\mathfrak{g}$ is non-abelian) ideal of $\mathfrak{g}$, hence must be the whole Lie algebra $\mathfrak{g}$ (by simplicity of $\mathfrak{g}$). □

Theorem 3.5 ([Kna02, Theorem 1.54])**.** *$\mathfrak{g}$ is semisimple if and only if it is a direct sum of simple Lie algebras, i.e. $\mathfrak{g} = \bigoplus_j \mathfrak{g}_j$ for some simple Lie algebras $\mathfrak{g}_j$. If so, any ideal of $\mathfrak{g}$ must be a direct sum of various $\mathfrak{g}_j$.*

Proof. If $\mathfrak{g}$ is semisimple, take any minimal ideal $\mathfrak{a}$ of $\mathfrak{g}$. Then by minimality, $\mathfrak{a}$ must be a simple Lie algebra (ideal of $\mathfrak{a}$ must be ideal of $\mathfrak{g}$). By Lemma 3.3, $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$. The Lie algebra $\mathfrak{a}^\perp$ has smaller dimension. Then by an induction of the dimension of $\mathfrak{g}$, the theorem follows.

Conversely, if $\mathfrak{g} = \bigoplus_j \mathfrak{g}_j$, then for any ideal $\mathfrak{a}$ of $\mathfrak{g}$, the projection of $\mathfrak{a}$ onto the j-th factor, denoted by $\mathfrak{a}_j$, must be an ideal of $\mathfrak{g}_j$ (since the projection map is a Lie algebra homomorphism). Hence $\mathfrak{a}_j = 0$ or $\mathfrak{a}_j = \mathfrak{g}_j$. In the latter case, as

$$\mathfrak{g}_j = [\mathfrak{g}_j, \mathfrak{g}_j] \subseteq [\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a},$$

we have a direct sum decomposition

$$\mathfrak{a} = \bigoplus_{\mathfrak{g}_j \subseteq \mathfrak{a}} \mathfrak{g}_j.$$

Thus $[\mathfrak{a}, \mathfrak{a}] = \bigoplus_{\mathfrak{g}_j \subseteq \mathfrak{a}} [\mathfrak{g}_j, \mathfrak{g}_j] = \mathfrak{a}$, hence $\mathfrak{a}$ cannot be solvable. This implies that $\mathfrak{g}$ is semisimple. □

Corollary 3.6. *If $\mathfrak{g}$ is semisimple, then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.*

Proof. Decompose $\mathfrak{g} = \oplus_j \mathfrak{g}_j$, then

$$[\mathfrak{g}, \mathfrak{g}] = \oplus_j [\mathfrak{g}_j, \mathfrak{g}_j] = \mathfrak{g}.$$

□

Corollary 3.7. *The homomorphic image of a semisimple Lie algebra is semisimple, if it is nonzero (Homework).*

Consider an algebraic extension K/F.

Lemma 3.8. *Let B^K be the Killing form of $\mathfrak{g}^K$. Then $B^K|_{\mathfrak{g}\times\mathfrak{g}} = B$. As a consequence, $\mathfrak{g}^K$ is semisimple if and only if $\mathfrak{g}$ is.*

Proof. Fix a basis $\mathcal{B}$ of $\mathfrak{g}$. Then $\mathcal{B}$ is a basis of $\mathfrak{g}^K$ over K. Thus, for any $X \in \mathfrak{g}$, the matrix $\mathrm{ad}(X)$ is the same for $\mathfrak{g}$ as it is for $\mathfrak{g}^K$. Recall in Proposition 2.24 of Lecture 01, $\mathrm{rad}(\mathfrak{g}^K) = (\mathrm{rad}\mathfrak{g})^K$. This also implies that

$$\mathfrak{g}^K \text{ is semisimple} \Leftrightarrow \mathfrak{g} \text{ is semisimple}.$$

□

The following is a special case of the previous lemma.

Corollary 3.9 ([Kna02, Corollary 1.53])**.** *Let $\mathfrak{g}$ be a complex Lie algebra and $\mathfrak{g}_0$ be a real form (i.e. $\mathfrak{g}_0$ is a real Lie subalgebra such that $\mathfrak{g}_0^{\mathbb{C}} = \mathfrak{g}$). Then $\mathfrak{g}$ is semisimple if and only if $\mathfrak{g}_0$ is.*

In the rest of this section, we only focus on real and complex Lie algebras.

Lemma 3.10. *Suppose that $\mathfrak{g}$ is a complex Lie algebra. Let $B_{\mathbb{R}}$ be the Killing form of $\mathfrak{g}$ when $\mathfrak{g}$ is regarded as a real Lie algebra. Then $B_{\mathbb{R}} = 2\mathrm{Re}(B)$.*

Proof. Let $\mathrm{Mat}_n(F)$ be the algebra of $n \times n$ matrices over a given field F. Then we have an obvious $\mathbb{R}$-algebra embedding

$$\iota : \mathrm{Mat}_n(\mathbb{C}) \to \mathrm{Mat}_{2n}(\mathbb{R}), \quad A + \mathbf{i}B \mapsto \begin{pmatrix} A & B \\ -B & A \end{pmatrix}.$$

Then obviously, $\mathrm{Tr}(\iota(Z)) = 2\mathrm{Tr}(\mathrm{Re}(Z)) = 2\mathrm{Re}(\mathrm{Tr}(Z))$. This implies that $B_{\mathbb{R}} = 2\mathrm{Re}(B)$. □

Proposition 3.11. *Let $\mathfrak{g}$ be a complex Lie algebra. Then $\mathfrak{g}$ is semisimple as a complex Lie algebra if and only if it is semisimple as a real Lie algebra.*

Proof. This is obvious from Lemma 3.10 and Cartan's criterion for semisimplicity (Theorem 2.5). □

Corollary 3.12. *Let $\mathfrak{g}$ be a complex simple Lie algebra. Then it is also simple as a real Lie algebra. (Homework)*

4. Weyl's Theorem on Completely Reducibility

We continue to assume that $\mathfrak{g}$ is a finite dimensional Lie algebra over a field F of characteristic zero.

Theorem 4.1 (Weyl, [Var84, Theorem 3.13.1])**.** *A finite dimensional representation of a semisimple Lie algebra over F is completely reducible.*

To the best of my knowledge, there are at least three strategies to prove Theorem 4.1. The analytic strategy, usually called Weyl's unitarian trick, only works over $F = \mathbb{C}$. One needs to find a compact real form of a complex semisimple Lie algebra, then average over a compact group by integration. Such a strategy can be dated back to the work of Hurwitz and Schur (appology for the absence of a reference, forget where I read this story). We will discuss this strategy in later lectures. One algebraic strategy requires knowledge of the structure of an irreducible finite-dimensional representation over $\overline{F}$. As suggested, this strategy only works over an algebraically closed field and cannot be presented before the full highest weight theory is established. This lecture notes follows the third strategy via the Lie algebra cohomology, as it requires less preliminaries and works for an arbitrary field of characteristic zero. We will discuss Lie algebra cohomology later in the course and prove Weyl's theorem then.

5. Reductive Lie Algberas

We continue to assume that $\mathfrak{g}$ is a finite-dimensional Lie algebra over F.

Definition 5.1. *The Lie algebra $\mathfrak{g}$ is said to be **reductive** if for each ideal $\mathfrak{a}$ in $\mathfrak{g}$, there exists an ideal $\mathfrak{b}$ in $\mathfrak{g}$ such that $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$.*

Definition 5.2. *Let V be a representation of $\mathfrak{g}$, not necessarily of finite dimensional. We say V is **completely reducible** if for any $\mathfrak{g}$-invariant subspace W of V, there exists a $\mathfrak{g}$-invariant subspace U such that*

$$V = W \oplus U.$$

Remark 5.3. *The definition amounts to say that the adjoint representation of $\mathfrak{g}$ on itself is completely reducible.*

Obviously, a semisimple Lie algebra must be reductive.

Proposition 5.4 ([Kna02, Corollary 1.56])**.** *If $\mathfrak{g}$ is reductive, then $\mathfrak{g} = \mathfrak{z}_{\mathfrak{g}} \oplus [\mathfrak{g}, \mathfrak{g}]$, where $\mathfrak{z}_{\mathfrak{g}}$ is the center and $[\mathfrak{g}, \mathfrak{g}]$ is semisimple.*

Proof. By an induction on the dimension of $\mathfrak{g}$, we can show that

$$\mathfrak{g} = (\bigoplus_j \mathfrak{a}_j) \oplus (\bigoplus_j \mathfrak{b}_j)$$

where $\dim \mathfrak{a}_j = 1$, $\mathfrak{b}_j$ are ideals of $\mathfrak{g}$ of dimension greater than one that contains no smaller ideals. Then $\mathfrak{b}_j$ are simple Lie algebras. This shows that $\mathfrak{z}_{\mathfrak{g}} = \bigoplus_j \mathfrak{a}_j$ and $[\mathfrak{g}, \mathfrak{g}] = \bigoplus \mathfrak{b}_j$. □

Remark 5.5. *Obviously, the above proposition implies that* $\mathrm{rad}(\mathfrak{g}) = \mathfrak{z}_{\mathfrak{g}}$. *In fact, a Lie algebra $\mathfrak{g}$ is reductive if and only if* $\mathrm{rad}(\mathfrak{g}) = \mathfrak{z}_{\mathfrak{g}}$. *Yet (to the best of my knowledge) it requires more preparation (such as Levi decomposition) to prove this claim. For details, see* [Var84, Theorem 3.16.3].

Proposition 5.6 ([Kna02, Proposition 1.59])**.** *Let $\mathfrak{g}$ be a real Lie subalgebra of $\mathfrak{gl}_n(\mathbb{R})$ or $\mathfrak{gl}_n(\mathbb{C})$ (regarded as a real Lie algebra). If $\mathfrak{g}$ is closed under conjugate transpose, then $\mathfrak{g}$ is reductive.*

Proof. For $X, Y \in \mathfrak{g}$, define $\langle X, Y\rangle = \mathrm{Re}(\mathrm{Tr}(X\overline{Y}^t))$. Then it is easy to check that $\langle \cdot\,,\,\cdot\rangle$ is symmetric, positive-definite, $\mathbb{R}$-bilinear form on $\mathfrak{g}$. Let $\mathfrak{a}$ be an ideal in $\mathfrak{g}$ and $\mathfrak{a}^\perp$ be the orthogonal complement of $\mathfrak{a}$ in $\mathfrak{g}$ (w.r.t $\langle \cdot\,,\,\cdot\rangle$). Then for any $X \in \mathfrak{a}^\perp, Y \in \mathfrak{g}, Z \in \mathfrak{a}$,

$$\langle [X, Y], Z\rangle = -\langle X, [\overline{Y}^t, Z]\rangle = 0.$$

This shows that $\mathfrak{a}^\perp$ is an ideal of $\mathfrak{g}$. □

The above proposition helps us to show the reductivity of the following Lie algebras.

Example 5.7. (1) *For $F = \mathbb{R}$ or $\mathbb{C}$, $\mathfrak{gl}_n(F), \mathfrak{sl}_n(F), \mathfrak{sp}_{2n}(F)$ are reductive.*
(2) *All $\mathfrak{so}(p, q)$, $\mathfrak{su}(p, q)$, $\mathfrak{u}(p, q)$ are reductive.*
(3) *All $\mathfrak{so}_n(\mathbb{C})$ are reductive.*

6. Proof of Cartan's Trace-Form Criteria

We prove Theorem 1.2 in this section. The proof we presented here is an adaptation of the argument in [Go-Wal09, Theorem 2.5.7].

Recall that if $\mathfrak{g}$ is solvable, then $\mathfrak{g} \neq [\mathfrak{g}, \mathfrak{g}]$ (otherwise the commutator series will never terminate). The condition that $\mathfrak{g} \neq [\mathfrak{g}, \mathfrak{g}]$ implies the existence of a codimensional ideal $\mathfrak{h}$ of $\mathfrak{h}$ (see the proof of Engel's theorem in Lecture 01). Also recall that $\mathfrak{g}$ is solvable if and only if $[\mathfrak{g}, \mathfrak{g}]$ is solvable (Proposition 2.11 in Lecture 01). This indicates a proof by induction on the dimension of $\mathfrak{g}$.

Lemma 6.1. *Assume that F is algebraically closed. Let $\mathfrak{h}$ be a codimensional one solvable Lie subalgebra of $\mathfrak{g}$. Then there exists $Y \in \mathfrak{g}, Y \notin \mathfrak{h}, \mu \in \mathfrak{h}^*$ such that $[X, Y] - \mu(X)Y \in \mathfrak{h}$ for all $X \in \mathfrak{h}$.*

Proof. Consider the adjoint representation of $\mathfrak{h}$ on $\mathfrak{g}$. As $\mathfrak{h}$ is an invariant space, it induces a representation $\pi : \mathfrak{h} \to \mathrm{End}_F(\mathfrak{g}/\mathfrak{h})$. Since F is algebraically closed, by Lie's theorem, the representation π admits a simultaneous eigenvector $\overline{Y} \in \mathfrak{g}/\mathfrak{h}$. This means that

$$\pi(X)\overline{Y} = \mu(X)\overline{Y}$$

for some $\mu \in \mathfrak{h}^*$ and all $X \in \mathfrak{h}$. Take any representative $Y \in \mathfrak{g}$ corresponds to $\overline{Y}$ under the natural quotient map. Then for all $X \in \mathfrak{h}$,

$$\mathrm{ad}(X)Y \equiv \mu(X)Y \mod \mathfrak{h}.$$

The lemma follows. □

Lemma 6.2. *Let F, $\mathfrak{h}$, $\mathfrak{g}$, Y, μ be as in Lemma 6.1. Assume that (π, W) is a finite dimensional representation of $\mathfrak{g}$. By Lie's theorem, there exists $\lambda \in \mathfrak{h}^*$ and $w_0 \in W$ such that $\pi(X)w_0 = \lambda(X)w_0$ for all $X \in \mathfrak{h}$. Define $w_k := \pi(Y)^k w_0$ and $W_{-1} = 0$, $W_k = span_F\{w_0, w_1, \cdots, w_k\}$. Then for any k, W_k is a subrepresentation of $\mathfrak{h}$. Moreover, if $W_k \neq W_{k-1}$, then $X \in \mathfrak{h}$ acts by the scalar $\lambda(X) + k\mu(X)$ on the one-dimensional quotient representation $W_k/W_{k-1} = F \cdot \overline{w_k}$.*

Proof. We prove by induction. The lemma is obvious for $k = 0$. Assume that the lemma holds for k, then for $k+1$,

$$\pi(X)w_{k+1} = \pi(X)\pi(Y)w_k = \pi(Y)\pi(X)w_k + \pi([X,Y])w_k \qquad \forall X \in \mathfrak{h}.$$

Note that

$$\pi([X,Y])w_k - \mu(X)w_{k+1} = \pi([X,Y] - \mu(X)Y)w_k \in \pi(\mathfrak{h})w_k \subseteq W_k.$$

By induction hypothesis,

$$\pi(X)w_k - (\lambda(X) + k\mu(X))w_k \in W_{k-1} \subseteq W_k.$$

Apply $\pi(Y)$ on both sides, we get

$$\pi(Y)\pi(X)w_k - (\lambda(X) + k\mu(X))w_{k+1} \in W_k.$$

Thus,

$$\pi(X)w_{k+1} - (\lambda(X) + (k+1)\mu(X))w_{k+1} \in W_k.$$

This finishes the proof of the lemma. □

Remark 6.3. *The idea behind the above proof comes from the natural filtration of the universal enveloping algebra (though we will not cover this topic shortly, students are encouraged to revisit this remark later in the course). Decompose $\mathfrak{g} = F \cdot Y + \mathfrak{h}$, then PBW theorem says that $W_k = U_k(\mathfrak{g}) \cdot w_0 = U_k(F \cdot Y) \cdot w_0$. Consider the adjoint action of $\mathfrak{g}$ on $U(\mathfrak{g})$. Each $U_k(\mathfrak{g})$ is $\mathfrak{g}$-stable (hence $\mathfrak{h}$-stable). Thus $(\mathrm{ad}(X)U_k(\mathfrak{g})) \cdot w_0 \subseteq U_k(\mathfrak{g}) \cdot w_0$ for all $X \in \mathfrak{h}$. This directly implies that $\mathfrak{h} \cdot W_k \subseteq W_k$. Yet, (at least to some extent) it is still inevitable to use induction to compute the eigenvalue of $X \in \mathfrak{h}$ on the quotient space W_k/W_{k-1}.*

Corollary 6.4. *Let F, $\mathfrak{h}$, $\mathfrak{g}$, Y, μ be as in Lemma 6.1. Assume that (π, W) is a finite dimensional irreducible representation of $\mathfrak{g}$. Then for any $X \in \mathfrak{h}$,*

$$\mathrm{Tr}(\pi(X)^2) = \mu(X)^2 \cdot \sum_{k=0}^{m} (k - \frac{m}{2})^2,$$

where $m = \dim W - 1$.

Proof. Define W_k as in Lemma 6.2. By irreducibility of W, we know that $m = \dim W - 1$ is the smallest integer such that $W_m = W_{m+1}$ and $W_m = W$. Then for all $X \in \mathfrak{h}$,

$$\mathrm{Tr}(\pi(X)) = \sum_{k=0}^{m} \lambda(X) + k\mu(X) = (m+1)(\lambda(X) + \frac{m}{2}\mu(X)).$$

The assumption $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$ implies that $\mathrm{Tr}(\pi(X)) = 0$ for all $X \in \mathfrak{h}$. Hence $\lambda = -\frac{m}{2}\mu$ (as linear functionals on $\mathfrak{h}$). Thus,

$$\mathrm{Tr}(\pi(X)^2) = \mu(X)^2 \cdot \sum_{k=0}^{m} (k - \frac{m}{2})^2.$$

□

Proof of Theorem 1.2 for algebraically closed field F. If F is algebraically closed, then by Lie's theorem, $\mathfrak{g}$ is triangularizable. Then for all $X \in \mathfrak{g}, Y \in [\mathfrak{g},\mathfrak{g}]$, $\mathrm{Tr}(XY) = 0$. Now we prove the converse by induction on the dimension of $\mathfrak{g}$. When $\dim \mathfrak{g} = 1$, $\mathfrak{g}$ is abelian, the theorem follows obviously. Now we assume that the theorem holds for all Lie algebra with dimension less than n. Note that for all $X \in \mathfrak{g}^1 = [\mathfrak{g},\mathfrak{g}] \subseteq \mathfrak{g}$ and $Y \in \mathfrak{g}^2 \subseteq \mathfrak{g}^1$, $\mathrm{Tr}(XY) = 0$. If $\mathfrak{g} \neq [\mathfrak{g},\mathfrak{g}]$, then by induction hypothesis, $[\mathfrak{g},\mathfrak{g}]$ is solvable. Thus $\mathfrak{g}$ is solvable (Proposition 2.9 in Lecture 01). Hence it remains to deduce a contradiction if $\mathfrak{g} = [\mathfrak{g},\mathfrak{g}]$.

Let $\mathfrak{h}$ be a maximal proper subalgebra. (Since any one dimensional space in $\mathfrak{g}$ is an abelian Lie subalgebra, a nonzero $\mathfrak{h}$ must exist.) As $\mathfrak{h}$ is solvable with a smaller dimension, by inductive hypothesis, $\mathfrak{h}$ is solvable. Consider the adjoint representation of $\mathfrak{h}$ on $\mathfrak{g}/\mathfrak{h}$. By Lie's theorem, this representation must have a one-dimensional invariant subspace $F \cdot \overline{Y}$, where $Y \in \mathfrak{g}$. This implies that $\mathfrak{h} + F \cdot Y$ is a Lie subalgebra of $\mathfrak{g}$. By the maximality of $\mathfrak{h}$, we must have $F \cdot \mathfrak{h} = \mathfrak{g}$, i.e. $\mathfrak{h}$ is a codimensional one Lie subalgebra of $\mathfrak{g}$. Let μ be as in Lemma 6.1.

Recall the Jordan-Hölder theorem: every finite-dimensional module V over an algebra admits a composition series. Let $\mathfrak{g}$ act on $V = F^n$ (by matrix multiplication). We call this representation π. Take a composition series

$$\{0\} = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots \subseteq V_r = V,$$

where each V_j is a subrepresentation of V and the quotient representation $W_j := V_j/V_{j-1}$ is irreducible. By Corollary 6.4, we find that the trace of $\pi(X)^2$ on the quotient representation W_j is

$$\mu(X)^2 \cdot \sum_{k=0}^{m_j} (k - \frac{1}{2}m_j)^2,$$

Hence

$$\mathrm{Tr}(\pi(X)^2) = \mu(X)^2 \cdot \sum_{j=1}^{r} \sum_{k=0}^{m_j} (k - \frac{1}{2}m_j)^2.$$

By the assumption in the theorem, $\mathrm{Tr}(\pi(X)^2) = 0$, (as $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$). Hence

$$\mu(X)^2 \cdot \sum_{j=1}^{r} \sum_{k=0}^{m_j} (k - \frac{1}{2}m_j)^2 = 0.$$

As $\mu \neq 0$, then all $\dim W_j = 1$ (i.e. $m_j = 0$). But this implies that $\mathfrak{g}$ is triangularizable, hence solvable. Then $[\mathfrak{g},\mathfrak{g}] \neq \mathfrak{g}$. We get a contradiction. □

Proof of Theorem 1.2, general case. Let $K = \overline{F}$, an algebraic closure of F. Note that $[\mathfrak{g},\mathfrak{g}]^K = [\mathfrak{g}^K, \mathfrak{g}^K]$ (Proposition 1.14 in Lecture 01). Also note that $\mathfrak{g}$ is solvable if and

only if $\mathfrak{g}^K$ is solvable (Proposition 2.9 in Lecture 01). Thus,

$$\begin{aligned}\mathfrak{g} \text{ is solvable } &\Leftrightarrow \mathfrak{g}^K \text{ is solvable}\\ &\Leftrightarrow \mathrm{Tr}(XY) = 0 \quad \forall X \in \mathfrak{g}^K, Y \in [\mathfrak{g}^K, \mathfrak{g}^K]\\ &\Leftrightarrow \mathrm{Tr}(XY) = 0 \quad \forall X \in \mathfrak{g}, Y \in [\mathfrak{g}, \mathfrak{g}].\end{aligned}$$

□

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Department of Mathematics, Zhejiang University, China
Email address: tianfangyangmath@zju.edu.cn